Bulk-Silicon Parent Qualification and Joint-Reduction Technical Report

Project Koios

Draft calculation report

Abstract

This report defines the calculation boundary for a prospective scalar-relativistic, non-spin-polarized, non-spin–orbit-coupled bulk-silicon parent, a compatible ten-spatial-orbital Wannier bridge, and a finite-support local-model hierarchy. Existing evidence is limited to an audited provisional direct bootstrap. No production parent, qualified Wannier representation, joint-fidelity result, or material-validation claim is established.

1 Evidence status

This is a publication-independent calculation report. It describes a prospective material calculation with an audited provisional bootstrap. It is not a completed production calculation, validated silicon Hamiltonian, qualified Wannier representation, model-selection result, or scientific validation. No new scientific calculator was executed for this report.

2 Scientific objective

The target is a pristine bulk-silicon parent represented consistently at three levels:

1. a first-principles plane-wave oracle $O_A(\mathbf{k})$;
2. a retained ten-spatial-orbital Wannier bridge $O_B(\mathbf{k})$; and
3. a finite-support sp^3s^* local-model hierarchy $O_C^{U,S}(\mathbf{k})$.

For retained frame $Q(\mathbf{k})$, alignment $U(\mathbf{k})$, support S, and real-space blocks $T_U(\mathbf{R})$, the intended relations are

$$O_B(\mathbf{k}) = Q(\mathbf{k})^\dagger O_A(\mathbf{k}) Q(\mathbf{k}), \tag{1}$$

$$O_C^{U,S}(\mathbf{k}) = \sum_{\mathbf{R}\in S} e^{i\mathbf{k}\cdot\mathbf{R}} T_U(\mathbf{R}). \tag{2}$$

A candidate is adequate only if the same model, support, alignment, basis, and energy reference satisfy both declared spectral and aligned-operator tolerances. Visual band agreement alone is insufficient.

3 External empirical development control

The calculation-local Vogl sp^3s^* control follows the empirical model of Vogl, Hjalmarson, and Dow [1] and contains the accepted complete Table 1 transcription: 16 compounds, 13 parameters per compound, and energies in eV. An independent implementation constructs the nearest-neighbor, spinless, non-SOC ten-band Bloch Hamiltonian in the ordered anion–cation basis $(s, p_x, p_y, p_z, s^*)_a \oplus (s, p_x, p_y, p_z, s^*)_c$. Its reduced Cartesian wavevector uses $\mathbf{k} = (2\pi/a_L)\mathbf{q}$.

Software and analytic checks cover the table inventory, silicon parameter row, structure factors at symmetry points, exact Hermitian assembly, the silicon Γ spectrum, reciprocal periodicity, and invalid-input rejection. This external empirical control is not derived from the provisional first-principles bootstrap or a Wannier bridge and does not freeze the candidate inventory, establish model adequacy, or qualify a material result.

4 Existing evidence

The historical direct bootstrap evaluated six wavefunction-cutoff candidates and four reciprocal-mesh candidates, with exact reuse of the C48 branch as logical K8. This produced nine unique self-consistent-field (SCF) and nine linked diagnostic non-self-consistent-field (NSCF) executions. The compact audit recomputed the required identities and observables and found the retained matrix complete and internally consistent.

Several adjacent comparisons meet the frozen finite-setting criteria. Human disposition selected C48/K8—48/192 Ry with a shifted $8 \times 8 \times 8$ mesh—as a provisional finite-setting parameter pair. This establishes neither an accepted production setting nor an infinite-basis error bound. The provisional geometry is not an accepted equilibrium lattice, the three diagnostic momenta do not locate the conduction valley, and printed fixed-point bands do not converge an effective mass.

Every invocation also emitted the same IEEE flag report. Instrumented diagnoses localized one event to external OpenCL discovery during startup and classified a separate cross-BLAS, cross-MPI divide-by-zero below `ZHEGVX` during initial-wavefunction subspace diagonalization. The effect on retained observables and the sources of the remaining reported flag classes are unresolved. Successful exit and `JOB DONE.` do not establish harmlessness. A mixed cutoff–mesh interaction check implied by C48/K8 would be new protected execution.

Exact values, hashes, criteria, and warning investigations remain at the historical locations catalogued in `calculations/Si/bulk/evidence-crosswalk.md`. They are referenced rather than copied or relabelled.

5 Required parent qualification

The production parent must freeze the physical branch, lattice, PBE/ONCV approximation, Quantum ESPRESSO executable, pseudopotential identity, occupations, solver settings, cutoff and mesh inventory, compact/native artifact boundary, and warning policy before execution. Qualification must keep energy, stress, fixed-point bands, gaps, valley identity, and curvature requirements distinct.

For each retained observable q, finite-setting agreement

$$|q_j - q_{j+1}| \leq \tau_q \tag{3}$$

demonstrates only stability across the tested pair. It does not imply a bound on $|q_j - q_\infty|$ without an additional reviewed error argument.

6 Required retained-space bridge

The Wannier bridge must retain projections, windows, centers, spreads, convergence status, reciprocal sewing, symmetry and orbital semantics, energy reference, and direct parent comparisons. At crossings or degeneracies, comparison must use spectral clusters or projectors rather than undocumented individual eigenvector labels.

Wannier90 may produce a candidate witness. Wannier90 convergence is not by itself a proof of the bridge contract. The tutorial-scale silicon Wannier extraction remains an integration precursor and is excluded from the production admissible sets.

7 Joint spectral/operator decision

For a frozen finite candidate inventory D, define

$$F(\tau_E, \tau_O) = \{d \in D : L_E(d) \le \tau_E \ \wedge \ L_O(d) \le \tau_O\}. \tag{4}$$

If this set is nonempty, the intended finite decision is

$$C^*(\tau_E, \tau_O) = \min_{d \in F(\tau_E, \tau_O)} C(d). \tag{5}$$

The same d must determine the model, support, alignment, and energy reference in both loss channels. Separate spectral and operator optima do not form a common witness.

The decision must preserve distinct numerical, subspace, alignment, interpolation, and support-truncation error channels. It must report one of the following:

- a common witness and the smallest adequate declared model;
- a reviewed separation result for an exhaustively evaluated finite inventory; or
- an explicit inconclusive outcome.

Failure of a local optimizer does not prove infeasibility.

8 Withheld analysis and residual attribution

Construction and withheld momentum sets must be disjoint by contract. Withheld analysis must report band, gap, valley, effective-mass, projector, represented-operator, and locality channels separately. Stable residuals may be attributed by onsite or hopping block, orbital semantics, symmetry channel, and support shell only when the representation and alignment contracts permit that interpretation.

Sensitivity analysis must include every prospectively declared parent, Wannier, stencil, alignment-map resolution, and threshold variation. Post-hoc sensitivity does not retroactively justify thresholds.

9 Verification and retention

A completed calculation requires independent verification of sources, schemas, identities, finite candidate coverage, objective reconstruction, threshold evaluation, and withheld results. Generated plots must consume verified records. A retained package must be no-overwrite and checksum-bound.

Formal proofs or certificates may establish exact implications for finite definitions and accepted interval premises. They do not certify Quantum ESPRESSO or Wannier90 execution, JSON, binary64 rounding, pseudopotential transferability, or physical validation.

10 Current gate and next decision

The active successor disposition permits explicitly labelled downstream development and exploratory analysis using provisional C48/K8 evidence. Production-parent and publication-grade acceptance remain pending an observable-level comparison through an independently identified established QE/toolchain path and the absent mixed cutoff–mesh interaction check. Capturing and replaying the exact first 8×8 subspace Hamiltonian and overlap matrices remains an optional focused diagnosis rather than a prerequisite for all downstream work. Every new QE invocation requires separate authorization. The detailed gates are recorded in `calculations/Si/bulk/development-plan.md` and the active decision in `calculations/Si/bulk/provisional-use-disposition.md`.

11 Nonclaims

This report does not establish production convergence, an infinite-basis result, experimental agreement, pseudopotential transferability, a qualified Wannier model, an adequate tight-binding model, or scientific validation. It authorizes no Quantum ESPRESSO or Wannier90 execution, evidence promotion, publication, or release.

A Supplemental material

A.1 Technical note: Vogl-control implementation provenance

The scientific definition and parameter authority for the empirical control are the published Vogl model and its Table 1 [1]. A separate development lineage was inspected for comparison: a public zinc-blende LCAO Gist by Mahdisadjadi [2] and the 2026 `SiliconLCAO_4sDOS-v3` notebook attributed within the notebook to Simon Phillpot [3]. The notebook states that it began from the Gist, corrected phase factors, and added the s^* orbitals, with modifications dated 29 April through 11 May 2026.

The calculation-local module `vogl1983_lcao_sp3sstar/model.py` was written independently from the published model equations and the accepted Table 1 transcription. No Gist or notebook source code was copied. Their role is limited to development provenance and numerical cross-checking; they do not supersede the primary source. The notebook agreed with 203 of the 208 Table 1 cells at four-decimal numerical normalization. The five disagreements were resolved in favor of the printed Vogl values. Neither the Gist nor the notebook exposed an explicit license in the inspected material, which is an additional reason to keep their role separate from the maintained implementation.

A.2 Technical note: LCAO and tetrahedral sp^3 construction

This summary follows the tight-binding treatments of Harrison, Chadi and Cohen, and Vogl et al. [4, 5, 1]. At one site, begin with the orthonormal orbitals $|s\rangle$, $|p_x\rangle$, $|p_y\rangle$, and $|p_z\rangle$,

$$\langle \alpha \mid \beta \rangle = \delta_{\alpha\beta}, \qquad \alpha, \beta \in \{s, p_x, p_y, p_z\}. \tag{6}$$

The four sp^3 hybrid orbitals are formed by combining the s, p_x, p_y, and p_z orbitals with equal magnitudes but different sign patterns:

$$\begin{pmatrix} |h_1\rangle \\ |h_2\rangle \\ |h_3\rangle \\ |h_4\rangle \end{pmatrix} = \frac{1}{2} \underbrace{\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}}_{H} \begin{pmatrix} |s\rangle \\ |p_x\rangle \\ |p_y\rangle \\ |p_z\rangle \end{pmatrix}. \tag{7}$$

The sign matrix H is a 4×4 Hadamard matrix. Its rows are mutually orthogonal and satisfy

$$HH^{\mathsf{T}} = 4I. \tag{8}$$

Consequently, $H/2$ defines an orthogonal change of basis that preserves norms and inner products. It transforms the orthonormal atomic-orbital basis into four orthonormal hybrid orbitals, $\langle h_m \mid h_n \rangle = \delta_{mn}$. Each row's sign pattern selects one of the $(1,1,1)$, $(1,\bar{1},\bar{1})$, $(\bar{1},1,\bar{1})$, or $(\bar{1},\bar{1},1)$ tetrahedral bond directions. Because this normalized Hadamard transform is symmetric and orthogonal, the same matrix also recovers the original s and p orbitals from the hybrid basis.

For a periodic solid, form Bloch sums of localized orbitals,

$$|\alpha b \mathbf{k}\rangle = \frac{1}{\sqrt{N}} \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot(\mathbf{R}+\boldsymbol{\nu}_b)} |\alpha b \mathbf{R}\rangle, \tag{9}$$

where b labels the anion or cation sublattice. Translation therefore enters a Hamiltonian matrix element through a phase $e^{i\mathbf{k}\cdot\mathbf{r}}$; reversing the bond gives the complex-conjugate phase required by Hermiticity. The LCAO approximation represents each symmetry-distinct localized matrix element by a constant: onsite E_s and E_p, and nearest-neighbor V_{ss}, V_{sp}, V_{xx}, and V_{xy}. In a heteropolar crystal, anion and cation onsite energies differ, and $V(s_a, p_c)$ and $V(s_c, p_a)$ are separate directional parameters. Onsite s–p matrix elements vanish by the site symmetry of the orthogonal basis; intersite s–p entries acquire signs from the odd parity of the p orbital.

For the four nearest-neighbor displacements

$$\mathbf{r}_{1,2,3,4} = \frac{a_L}{4}\big\{(1,1,1), (1,-1,-1), (-1,1,-1), (-1,-1,1)\big\}, \tag{10}$$

write $e_n = e^{i\mathbf{k}\cdot\mathbf{r}_n}$. The four structure factors are

$$g_0 = \tfrac{1}{4}(e_1 + e_2 + e_3 + e_4), \qquad g_1 = \tfrac{1}{4}(e_1 + e_2 - e_3 - e_4), \tag{11}$$

$$g_2 = \tfrac{1}{4}(e_1 - e_2 + e_3 - e_4), \qquad g_3 = \tfrac{1}{4}(e_1 - e_2 - e_3 + e_4). \tag{12}$$

Thus representative Bloch-matrix entries are $g_0 V(s,s)$ for s–s coupling and $g_j V(s_a, p_{j,c})$ for s–p_j coupling, with the reverse entries fixed by the Hermitian conjugate and the orbital-direction sign convention. The sp^3 construction produces an 8×8 Hamiltonian for two four-orbital sites. Vogl's sp^3s^* model adds one excited s^* orbital per site and only the declared nearest-neighbor s^*–p couplings, producing the maintained 10×10 block form

$$H(\mathbf{k}) = \begin{pmatrix} D_a & T(\mathbf{k}) \\ T(\mathbf{k})^\dagger & D_c \end{pmatrix}, \qquad D_b = \operatorname{diag}\big(E(s,b), E(p,b), E(p,b), E(p,b), E(s^*,b)\big). \tag{13}$$

The Table 1 transfer parameters determine $T(\mathbf{k})$; the s^*–s and s^*–s^* transfers are structural zeros in this model.

References

[1] P. Vogl, H. P. Hjalmarson, and J. D. Dow. A semi-empirical tight-binding theory of the electronic structure of semiconductors. *Journal of Physics and Chemistry of Solids*, 44(5):365–378, 1983.

[2] Mahdisadjadi. Zinc-blende LCAO. Development Gist. Inspected 22 October 2026.

[3] Simon Phillpot. SiliconLCAO_4sDOS-v3, 2026. Development notebook; modifications recorded as 29 April–11 May 2026; inspected read-only 22 October 2026.

[4] Walter A. Harrison. *Electronic Structure and the Properties of Solids: The Physics of the Chemical Bond*. W. H. Freeman and Company, San Francisco, 1980.

[5] D. J. Chadi and M. L. Cohen. Tight-binding calculations of the valence bands of diamond and zincblende crystals. *physica status solidi (b)*, 68(1):405–419, March 1975.